

Day 5 Assignment – Data Visualization, AI Introduction & ML Foundations

Level 1 – Basic

Task 1: Chart-to-Purpose Matching

- 1 → Pie Chart
- 2 → Line Plot
- 3 → Bar Plot
- 4 → Histogram

Justification:

Goal 2: A line plot is best because it shows how values change over time.

Goal 4: A histogram groups heights into ranges and shows the number of students in each range.

Task 2: True/False

- 1. True – A label is the value a model tries to predict.
- 2. False – Deep Learning is a subset of Machine Learning.
- 3. False – Pie charts are not suitable for showing changes over time.
- 4. True – `plt.title()` adds a title above the chart.

Task 3: Missing Lines

```
plt.xlabel("X Axis")  
plt.ylabel("Y Axis")
```

Level 2 – Intermediate

Task 4: Read a Dataset and Split It

Features: distance_km, package_kg, traffic_level

Label: delivery_min

traffic_level is categorical data and cannot be used directly by most ML models. It must be converted into numbers using encoding techniques.

If there are 200 rows, I would use approximately 80% for training and 20% for testing. The testing data is kept separate so that the model can be evaluated on unseen data.

Task 5: Plan Two Visualisations

Average price by brand:

Chart: Bar Chart

X-axis: Brand

Y-axis: Average Price

Battery capacity vs Rating:

Chart: Scatter Plot

Positive relationship: As battery_mah increases, rating also tends to increase.

Key Matplotlib lines:

```
plt.scatter(df["battery_mah"], df["rating"])  
plt.xlabel("Battery Capacity")  
plt.ylabel("Rating")  
plt.title("Battery Capacity vs Rating")
```

Task 6: Classify the Technology

1. Birthday poem generator → Generative AI (creates new content).
2. Automatic transaction block rule → Rule-Based System (uses fixed rules).
3. Face recognition using neural network → Deep Learning.
4. Loan default prediction → Machine Learning (learns patterns from past data).

Level 3 – Challenge

Task 7: Design a Mini ML Plan

Features:

- Day of week
- Number of students present
- Weather conditions
- Festival/Event indicator

Label:

- Number of food plates required

Workflow:

1. Collect historical canteen data.
2. Clean missing or incorrect values.
3. Select features and label.
4. Split data into training and testing sets.
5. Train a machine learning model.
6. Evaluate model performance.
7. Generate daily predictions.
8. Use predictions to plan food preparation.

Last week's data may not perfectly predict a festival day because attendance and food demand can change significantly during special events. Testing on unseen data helps ensure the model can generalize to situations it has not encountered before.

Task 8: Interpret and Improve a Chart

A pie chart is not suitable because website visitors change hour by hour and pie charts do not effectively show trends over time.

A line chart would be a better choice because it clearly displays changes in visitor count throughout the day.

I would add grid lines using `plt.grid(True)`. Grid lines make it easier to compare visitor counts across different hours and identify peak traffic periods.